



COMPETENCY STANDARD

FOR

PNEUMATIC & HYDRAULIC SYSTEM OPERATION AND MAINTENANCE

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)

Finance Division, Ministry of Finance

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The Competency Standards for **Pneumatic & Hydraulic System Operation and Maintenance** is a document for developing curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:

Generac Competency (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-PHS-01-G	Apply Occupational Health and Safety (OHS) Practices in Workplace	1. Identifying OHS policies and procedures 2. Practicing personal health and safety procedures 3. Reporting hazards and risks 4. Responding to emergencies.	10
SICIP-LE-PHS-02-G	Operate a Self-Directed Team	1. Identifying team goals and work processes 2. Communicating and cooperating with team members 3. Working and solving problems as a team member	10
Total Hours			20

Sector Specific Competency (10 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-PHS-01-S	Apply Green Practices in Light Engineering Sector	1. Interpreting green concepts 2. Minimizing resources in the workplace 3. Implementing waste management practices	10
Total Hours			10

Occupation Specific Competency (270 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-PHS-01-O	Understand Concept of Pneumatic Control System	1. Identify components of pneumatic system 2. Interpret pneumatic circuit diagrams. 3. Demonstrate safe practice in handling pneumatic system	30
SICIP-LE-PHS-02-O	Use Electro-Pneumatic Control Application	1. Identify electro- pneumatic control circuits 2. Interpret logical control functions 3. Assemble electro-pneumatic circuits 4. Simulate and test Electro-pneumatic system 5. Apply electro- pneumatic control application circuits in training board	35
SICIP-LE-PHS-03-O	Perform PLC And HMI Programming	1. Identify PLC components 2. Identify HMI components 3. Configure PLC, HMI and PC 4. Develop PLC program for control system 5. Visualize pneumatic and hydraulic system on HMI screen	50
SICIP-LE-PHS-04-O	Perform Advanced Electro-Pneumatic Control Application	1. Identify components of PLC based electro- pneumatic control system 2. Interpret and develop PLC ladder logic for electro-pneumatic applications 3. Interface electro-pneumatic components with PLC 4. Program PLC for sequential control	40
SICIP-LE-PHS-05-O	Understand Concept of Hydraulic	1. Identify components of hydraulic system 2. Interpret hydraulic circuit	30

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Control System	diagrams. 3. Demonstrate safe practice in handling hydraulic system	
SICIP-LE-PHS-06-O	Use Electro-Hydraulic Control Application	1. Identify electro- hydraulic control circuits 2. Interpret logical control functions 3. Assemble electro- hydraulic circuits 4. Simulate and test electro- hydraulic system 5. Apply electro- hydraulic control application circuits in training board	30
SICIP-LE-PHS-07-O	Perform Advanced Electro-Hydraulic Control Application	1. Identify components of PLC based electro-hydraulic control system 2. Interpret and develop PLC ladder logic for electro- hydraulic applications 3. Interface electro-hydraulic components with PLC 4. Program PLC for sequential control	35
SICIP-LE-PHS-08-O	Troubleshoot And Maintain Pneumatic and Hydraulic System Operations	1. Inspect pneumatic and hydraulic system components 2. Diagnose faults in pneumatic and hydraulic systems 3. Perform maintenance on pneumatic and hydraulic systems 4. Test and verify system operation	20
Total Hours			0

A. The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-PHS-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variable

Variable	Range (May include but not limited to):
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices.
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	1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures. 1.4 Reported incidents arising from hazards and risks to authority. 1.5 Implemented plans and procedures to respond emergency. 1.6 Applied basic first aid procedure.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Unit of Competency:	Nominal Duration:	Unit Code:
OPERATE IN A SELF-DIRECTED TEAM	10 hrs.	SICIP-LE-PHS-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range (May Include but not limited to):
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued Identifying current and potential problems

	faced by the team 2.9 Identifying solutions to the problem 2.10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

2. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
3. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
4. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

B. The Sector Specific Competencies

Unit of Competency:	Nominal	Unit Code:
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APPLY GREEN PRACTICES IN LIGHT ENGINEERING SECTOR	Duration: 10 hrs.	SICIP-LE-PHS-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in light engineering sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Basic concepts of green</u> are understood. 1.2 Energy efficiency principles are interpreted. 1.3 <u>Sustainable materials</u> are identified. 1.4 Pollution control methods are understood. 1.5 <u>Environmental certifications and standards</u> are identified. 1.6 <u>Green supply chains</u> are interpreted to minimize environmental impact.
2. Minimize resources use in the workplace	2.1 Resource consumption is identified in the light engineering sector. 2.2 <u>Energy-efficient equipment</u> is identified and maintained. 2.3 Materials are sourced ensuring that sustainable and recyclable materials are prioritized. 2.4 Water usage is minimized through efficient management. 2.5 Production processes are optimized. 2.6 <u>Green technologies and innovations</u> are implemented in the workplace.
3. Implement waste management practices	3.1 <u>Waste types</u> identified in the light engineering sector. 3.2 Waste segregation practices are interpreted. 3.3 Recycling programs are established through external recycling facilities. 3.4 <u>Waste reduction strategies</u> are understood. 3.5 Safe disposal practices are described. 3.6 <u>Waste-to-energy solutions</u> are explored to reduce landfill contributions.

Range of Variables

Variable	Range
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	(May include but not limited to):
1. Basic concepts of green	1.1 Principles of Green Practices 1.2 Sources of Environmental Impacts 1.3 Resource Utilization 1.4 Sustainable Alternatives 1.5 Waste Management Practices 1.6 Green Habits at Work 1.7 Compliance and Regulations
2. Sustainable materials	2.1 Recycled Metals 2.2 Biodegradable Lubricants and Fluids 2.3 Composite Materials with Low Environmental Impact 2.4 Low-VOC Paints and Coatings 2.5 Eco-friendly Packaging Materials 2.6 Durable and Repairable Components
3. Environmental certifications and standards	3.1 ISO 14001: Environmental Management System (EMS) 3.2 LEED Certification (Leadership in Energy and Environmental Design) 3.3 RoHS (Restriction of Hazardous Substances) 3.4 REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) 3.5 Energy Star Certification 3.6 Eco-Labeling Standards 3.7 Green Building Codes & Local Environmental Compliance
4. Green supply chains	4.1 Environmentally Responsible Sourcing 4.2 Eco-Friendly Transportation and Logistics 4.3 Supplier Environmental Compliance 4.4 Sustainable Packaging and Distribution
5. Energy-efficient equipment	5.1 High-Efficiency Electric Motors (IE2/IE3/IE4 Rated) 5.2 Variable Frequency Drives (VFDs) 5.3 Energy-Efficient LED Lighting Systems 5.4 Energy-Star Rated Industrial Equipment 5.5 Heat Recovery and Insulation Systems
6. Green technologies and innovations	6.1 Renewable Energy Integration 6.2 Additive Manufacturing (3D Printing) 6.3 Smart Manufacturing Systems (Industry 4.0) 6.4 Waste-to-Energy Technologies 6.5 Low-Emission Welding and Cutting Equipment 6.6 Closed-Loop Cooling and Lubrication Systems 6.7 Eco-Innovation in Product Design
7. Waste types	7.1 Metal Waste 7.2 Plastic and Packaging Waste 7.3 Hazardous Waste 7.4 Electronic Waste (E-Waste) 7.5 Dust and Particulate Waste

	7.6 Wood and Pallet Waste
8. Waste reduction strategies	8.1 5S Workplace Organization 8.2 Lean Manufacturing Practices 8.3 Material Optimization and Nesting Techniques 8.4 Reusing and Repurposing Waste
9. Waste-to-energy solutions	9.1 Thermal Conversion (Incineration with Energy Recovery) 9.2 Gasification and Pyrolysis 9.3 Biomass Boilers Using Organic Waste 9.4 Use of Waste Oils as Alternative Fuel 9.5 Waste Heat Recovery Systems (WHRS) 9.6 On-Site Micro Energy Generation Units

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of pollution. 1.2 Pollution control technologies 1.3 Legal and environmental frameworks: overview of national environmental policies and local environmental compliance regulations. 1.4 Importance of ISO 14001 and other EMS (Environmental Management System) standards. 1.5 Classification of sustainable, recyclable, and hazardous materials used in production. 1.6 Strategies for minimizing water and energy use during material processing. 1.7 Waste hierarchy: reduce, reuse, recycle, recover, and disposal (5Rs). 1.8 Industrial waste-to-energy concepts.
2. Underpinning Skills	2.1 Measuring and analyzing energy and water consumption using meters. 2.2 Identifying and operating energy-efficient machinery. 2.3 Monitoring and logging resource consumption data. 2.4 Adjusting machine settings or workflow layouts. 2.5 Applying green inventory practices. 2.6 Segregating waste effectively into recyclable, hazardous, and general categories. 2.7 Implementing basic pollution control activities.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors

	in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

C. OCCUPATION SPECIFIC COMPETENCIES (270 HRS.)

Unit of Competency: UNDERSTAND CONCEPT OF PNEUMATIC CONTROL SYSTEM	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LE-PHS-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand concept of pneumatic control system. It specifically includes the tasks of identifying components of pneumatic system, interpreting pneumatic circuit diagrams and demonstrating safe practice in handling pneumatic system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Components of Pneumatic System	1.1 <u>Major pneumatic components</u> are recognized. 1.2 Functions of Major pneumatic components are described. 1.3 Symbols of pneumatic components are interpreted. 1.4 Input, output ports of <u>valves</u> and <u>actuators</u> are located and their purpose are explained. 1.5 Components are differentiated by operational type (manual, mechanical, solenoid)
2. Interpret Pneumatic Circuit Diagrams	2.1 Components of pneumatic circuit are identified. 2.2 Functions of individual pneumatic components in the diagram are explained. 2.3 The flow of compressed air within the circuit is traced and adjusted. 2.4 Relationships between different circuit components and their operational sequence are interpreted. 2.5 Troubleshooting points are identified based on the circuit diagram interpretation.
3. Demonstrate Safe Practice in Handling Pneumatic System	3.1 Safety procedures and guidelines are followed consistently during handling of pneumatic system elements. 3.2 Pneumatic circuit simulation is demonstrated. 3.3 Pneumatic application is demonstrated on training board. 3.4 Pneumatic components and tools are inspected for defects or hazards before use. 3.5 Pneumatic system elements are handled and operated

	according to safety standards and manufacturer instructions. 3.6 Potential risks and hazards are identified and controlled effectively to prevent accidents. 3.7 Unsafe conditions or practices are reported promptly to the relevant authority. 3.8 Safe shutdown and isolation procedures of pneumatic systems are carried out correctly.
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Range of Variables

Variable	Range (May includes but not limited to):
1. Major Pneumatic Components	1.1 Air compressor 1.2 Air reservoir 1.3 FRL (Filter- Regulator- Lubricator) Unit 1.4 Actuators 1.5 DCV (Directional control Valve) 1.6 FCV (Flow control Valve)
2. Valves	2.1 2/2, 3/2, 4/2, 5/2way DCV 2.2 1 way and 2way flow control Valve 2.3 Normally open and normally closed push Valve.
3. Actuator	3.1 Single acting cylinder 3.2 Double acting cylinder
4. Potential Risks and Hazards	4.1 High Pressure Release 4.2 Component Failure 4.3 Noise Exposure 4.4 Pinch/Crush Injuries 4.5 Incorrect Connections 4.6 Electrical Hazards

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic principles and functions of pneumatic components. 1.2 Standard symbols for pneumatic components. 1.3 Purposes of common pneumatic components. 1.4 Understand circuit diagrams and principles of pneumatic control circuits. 1.5 Functional relationships between circuit components. 1.6 Methods of troubleshooting circuit diagrams. 1.7 Hazards and risks associated with pneumatic systems. 1.8 Safe handling, installation, and maintenance procedures
2. Underpinning Skills	2.1 Identifying and distinguishing pneumatic components visually. 2.2 Applying knowledge of component function to practical identification.

	2.3 Demonstrating simulation of pneumatic circuit. 2.4 Demonstrating pneumatic application on training board. 2.5 Tracing flow paths and control logic from diagrams. 2.6 Operating circuit components and identifying troubleshooting points and faults. 2.7 Implementing safe manipulation and assembly of pneumatic components. 2.8 Identifying and mitigating of hazards. 2.9 Executing emergency shutdown and isolation procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified components of pneumatic system 1.2 interpreted pneumatic circuit diagrams. 1.3 demonstrated safe practice in handling pneumatic system
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: USE ELECTRO-PNEUMATIC CONTROL APPLICATION	Nominal Duration: 35 Hrs.	Unit Code: SICIP-LE-PHS-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to use electro- pneumatic control application. It specifically includes the tasks of identifying electro-pneumatic control circuits, interpreting logical control functions, assembling electro-pneumatic circuits, simulating and testing electro-pneumatic system and applying electro-pneumatic control application circuits in training board.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Electro-Pneumatic Control Circuits	1.1 Standard symbols for electro-pneumatic components are recognized and identified correctly. 1.2 Functions of each electro-pneumatic element within the circuit are explained accurately. 1.3 Electrical and pneumatic parts of the circuit are distinguished and categorized appropriately. 1.4 Connections between electrical and pneumatic components are interpreted. 1.5 The functions of <u>sensors</u> , actuators and <u>valves</u> in the circuit are described. 1.6 Circuit elements are matched to their respective control functions within the system.
2. Interpret Logical Control Functions	2.1 Relay-based logical operators and functions used in control systems are identified and described. 2.2 Input and output relationships within the logical control system are analyzed and explained. 2.3 Logical sequences and conditions of cylinder movement are demonstrated correctly based on the control diagram or description. 2.4 The effect of each logical function on actuators is determined and articulated. 2.5 Faults or irregularities in sequence control are detected through analysis. 2.6 The overall sequential control logic is interpreted to predict system behavior under different scenarios.
3. Assemble Electro-Pneumatic Circuits	3.1 Required electro-pneumatic components are selected and arranged within FluidSIM software according to the circuit design specifications. 3.2 Circuit diagrams and simulation instructions are

	interpreted accurately before assembly. 3.3 Electro-pneumatic components are connected correctly in FluidSIM following the circuit diagram and operational logic. 3.4 Electrical connections and pneumatic lines are configured and linked properly using FluidSIM tools.
4. Simulate and Test Electro-Pneumatic System	4.1 Simulation settings and parameters in FluidSIM are configured to match system operational requirements. 4.2 The assembled electro-pneumatic circuit is simulated in FluidSIM to replicate real-world conditions accurately. 4.3 System responses and performance data are monitored and recorded during the FluidSIM simulation. 4.4 Any faults or errors detected during the FluidSIM simulation are identified and documented. 4.5 Necessary corrections and adjustments are made within FluidSIM to address identified issues.
5. Apply Electro-Pneumatic Control Application Circuits in Training Board	5.1 Electro-pneumatic control circuits for two-cylinder (A and B) <u>sequences</u> are made based on result of the simulations. 5.2 Cylinders A and B along with corresponding sensors, are correctly identified and connected to the control circuit on the training board. 5.3 Pneumatic tubing and electrical wiring are connected according to the designed sequence control diagrams involving two cylinders. 5.4 Sensor inputs and cylinder movements are monitored to ensure synchronization and adherence to the intended sequence. 5.5 Faults or discrepancies in cylinder sequences or sensor feedback are identified and corrected. 5.6 Safety and operational procedures are followed throughout the setup, testing, and shutdown processes of the training board.

Range of Variables

Variable	Range (May includes but not limited to):
1. Sensors	1.1 Pneumatic Proximity Sensor 1.2 Magnetic Proximity Sensor 1.3 Capacitive Proximity 1.4 Optical Proximity
2. Valves	2.1 3/2way solenoid valve 2.2 5/2way solenoid valve
3. Sequences	3.1 A+A-B+B-

	3.2 A+B+A-B-
	3.3 A+B+B-A-

Curricular Content Guide

1. Underpinning Knowledge	1.1 Standard symbols used in electro-pneumatic circuit diagrams. 1.2 Functions of key electro-pneumatic components. 1.3 Electrical and pneumatic parts of the circuit. 1.4 Fundamentals of logic operations and their implementation in control systems. 1.5 Use of relay logic in electro-pneumatic control. 1.6 Assembly procedures and best practices for electro-pneumatic hardware. 1.7 Simulation software tools of FluidSIM and their features. 1.8 Testing procedures and performance criteria for electro-pneumatic systems. 1.9 Fault detection and corrective actions during testing. 1.10 Configuration and setup of training boards for electro-pneumatic circuits. 1.11 Sensor and actuator integration on training platforms.
2. Underpinning Skills	2.1 Interpreting electro-pneumatic circuit diagrams. 2.2 Recognizing electrical and pneumatic parts of the circuit. 2.3 Analysing and understanding logic control sequences. 2.4 Translating logical functions into practical control applications. 2.5 Diagnosing and correcting logic faults. 2.6 Handling, connecting, and securing electro-pneumatic components. 2.7 Wiring electrical connections and routing pneumatic lines. 2.8 Operating simulation software to model electro-pneumatic circuits. 2.9 Setting up test parameters and monitoring system performance. 2.10 Configure and operate training board setups for electro-pneumatic control. 2.11 Adjusting control sequences for practical applications.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in

	the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified electro- pneumatic control circuits 1.2 interpreted logical control functions 1.3 assembled electro- pneumatic circuits 1.4 simulated and test Electro- pneumatic system 1.5 applied electro- pneumatic control application circuits in training board
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PLC AND HMI PROGRAMMING	Nominal Duration: 50 Hrs.	Unit Code: SICIP-LE-PHS-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform PLC and HMI programming. It specifically includes the tasks of identifying PLC components, identifying HMI components, configuring PLC, HMI and PC, developing PLC program for control system and visualizing pneumatic and hydraulic system on HMI screen		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify PLC Components	1.1 <u>PLC hardware components</u> are identified and labeled correctly according to manufacturer specifications. 1.2 The function of each PLC component, including CPU, power supply, input/output modules, communication ports, and programming devices, is described accurately. 1.3 PLC components are distinguished based on type, model, and functionality. 1.4 Safety features and protective devices integrated within the PLC system are identified. 1.5 Relevant documentation, such as PLC manuals and datasheets, is consulted and interpreted correctly for component identification.
2. Identify HMI Components	2.1 Human-Machine Interface (HMI) hardware components are identified and labeled correctly according to manufacturer specifications. 2.2 The function of each HMI component, including display screens, input devices (touchscreens, buttons), communication modules, and power supply units, is described accurately. 2.3 HMI components are distinguished based on type, model, and interface capabilities. 2.4 Safety features and protective devices within the HMI system are identified. 2.5 Relevant HMI manuals, datasheets, and technical documentation are consulted and interpreted correctly for component identification.
3. Configure PLC HMI and PC	3.1 PLC hardware and software configurations are set up according to system specifications and application requirements. 3.2 HMI devices are configured to communicate effectively

	with the PLC using appropriate protocols and settings. 3.3 Communication parameters between PLC, HMI, and PC are established and tested for reliability. 3.4 User interface elements on the HMI are designed and configured to meet operational needs. 3.5 Configured systems are tested to verify correct operation and communication between PLC, HMI, and PC. 3.6 Troubleshooting is conducted to resolve any configuration or communication issues identified during testing.
4. Develop PLC Program for Control System	4.1 PLC program logic is designed following standard programming practices and safety guidelines. 4.2 Basic PLC instructions such as timers, counters, contacts, coils, and data handling commands are applied appropriately in the program. 4.3 Logical programming constructs including AND, NAND, OR, NOR, NOT, XOR, XNOR operations are implemented correctly within the PLC code. 4.4 Program code is written using appropriate PLC Ladder programming. 4.5 The PLC program is simulated or tested in a controlled environment to verify functional correctness. 4.6 The completed PLC program is downloaded to the PLC hardware and configured for operation. 4.7 Program performance is monitored during operation to ensure reliability and compliance with system requirements.
5. Visualize Pneumatic and Hydraulic System on HMI Screen	5.1 HMI screen is designed for pneumatic components. 5.2 HMI screen is designed for hydraulic components. 5.3 Cylinder sequence is visualized for pneumatic system. 5.4 Cylinder sequence is visualized for hydraulic system.

Range of Variables

Variable	Range (May includes but not limited to):
1. PLC Hardware Components	1.1 CPU (Central Processing Unit) 1.2 Power Supply Module 1.3 Digital Input Modules 1.4 Digital Output Modules 1.5 Analog Input Modules 1.6 Communication Modules (e.g., Ethernet, Profinet) 1.7 Signal Conditioners 1.8 Mounting Rack or Baseplate 1.9 Programming Device (e.g., PC or Handheld)

	Programmer) 1.10 Memory Cards / Storage Devices 1.11 Terminal Blocks and Wiring Connectors
2. Protocols	2.1 Profinet 2.2 Ethernet

Curricular Content Guide

1. Underpinning Knowledge	1.1 Overview of PLC systems and architecture. 1.2 Functions of PLC hardware components. 1.3 Types of input and output modules and their uses. 1.4 Overview of Human-Machine Interface (HMI) systems and applications. 1.5 Functions of each HMI component. 1.6 Software tools for PLC programming and HMI configuration. 1.7 setup and protocol configuration between PLC, HMI, and PC. 1.8 PLC programming languages and standards. 1.9 Logical programming constructs: AND, NAND, OR, NOR, NOT, XOR, XNOR operations. 1.10 Testing and commissioning procedures. 1.11 Principles of pneumatic and hydraulic systems relevant to visualization. 1.12 Types and functions of sensors used for data acquisition in pneumatic and hydraulic systems. 1.13 Symbols and animations used for pneumatic and hydraulic components on HMI screens 1.14 Operation principles of pneumatic and hydraulic cylinders. 1.15 Cylinder movement sequences and their control methods.
2. Underpinning Skills	2.1 Identifying of component functions in practical applications. 2.2 Identifying various HMI devices and their interfaces. 2.3 Interpreting HMI specifications and technical documentation. 2.4 Using PLC and HMI software configuration tools. 2.5 Configuring network settings and troubleshoot connectivity. 2.6 Writing clear and efficient PLC code using relevant ladder diagram. 2.7 Developing logical control sequences and implement control algorithms.

	2.8 Integrating sensor data and PLC signals into HMI visualization screens. 2.9 Designing clear graphical interfaces showing cylinder movement stages.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere and honest to duties. 3.4 Environmental concerns. 3.5 Eagerness to learn. 3.6 Tidiness and timeliness. 3.7 Respect for rights of peers and seniors in the workplace. 3.8 Communication with peers, subordinates and seniors in the workplace.
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified PLC components 1.2 identified HMI components 1.3 configured PLC, HMI and PC 1.4 developed PLC program for control system 1.5 visualized pneumatic and hydraulic system on HMI screen
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	Nominal	Unit Code:
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PERFORM ADVANCED ELECTRO-PNEUMATIC CONTROL APPLICATION	Duration: 40 Hrs.	SICIP-LE-PHS-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform advanced electro-pneumatic control application. It specifically includes the tasks of identifying components of PLC based electro- pneumatic control system, interpreting and developing PLC ladder logic for electro-pneumatic applications, interfacing electro-pneumatic components with PLC and programing PLC for sequential control.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Electro-Pneumatic Control System Components	1.1 <u>Electro-pneumatic system components</u> are recognized and described accurately. 1.2 <u>Integration interfaces</u> between PLC and electro-pneumatic components are identified and explained. 1.3 <u>Signal types</u> used for PLC to electro-pneumatic system interaction are recognized. 1.4 <u>Protective components</u> within the control system are identified and their functions explained. 1.5 <u>Technical documentations</u> are consulted and interpreted.
2. Interpret and Develop PLC Ladder Logic for Electro-Pneumatic Applications	2.1 PLC ladder logic diagrams are analyzed and interpreted to understand the required control sequences. 2.2 Electro-pneumatic system requirements are translated into ladder logic functions. 2.3 Ladder diagram is developed using <u>appropriate devices</u> . 2.4 The ladder logic program is simulated or tested to verify correct operation of the electro-pneumatic control sequences. 2.5 Programming errors and logic faults are detected and corrected. 2.6 Ladder logic programs are downloaded to the PLC hardware and integrated with electro-pneumatic system.
3. Interface Electro-Pneumatic Components with PLC	3.1 Electro-pneumatic components are identified and prepared for PLC interfacing. 3.2 Electrical connections between PLC modules and electro-pneumatic devices are made. 3.3 Signal types from electro-pneumatic components are matched with PLC input/output specifications.

	3.4 Communication protocols for PLC and electro-pneumatic components are recognized and implemented. 3.5 Interface wiring and connections are tested for continuity, signal integrity and operation. 3.6 Integration of PLC and electro-pneumatic components is verified.
4. Program PLC for Sequential Control	4.1 Sequential control requirements for 3 cylinders (A, B & C) operations are analyzed and documented based on job specifications. 4.2 PLC ladder logic is used to develop control sequences for 3-cylinder operations. 4.3 Control logic is expanded to incorporate 3-cylinder sequential operations, managing synchronization and interlocking between cylinders. 4.4 Timers, counters, and sensor inputs are integrated to manage the sequence timing and cylinder position feedback. 4.5 The final PLC program is downloaded to the hardware and tested. 4.6 System operation is monitored to ensure the sequential control of cylinders is performed safely, reliably, and according to design.

Range of Variables

Variable	Range (May includes but not limited to):
1. Electro-Pneumatic System Components	1.1 Single acting cylinder 1.2 Double acting cylinder 1.3 Solenoid valves 1.4 Sensors 1.4.1 Capacitive Proximity sensor 1.4.2 Active Proximity sensor 1.4.3 Magnetic Proximity sensor 1.4.4 Optical Proximity sensor 1.4.5 Analog Proximity sensor
2. Integration Interfaces	2.1 Digital I/O (Input /Output) 2.2 Analog I/O 2.3 Communications Protocols 2.4 Profinet 2.5 Ethernet
3. Protective Components	3.1 Pressure Relief Valve 3.2 Pressure Regulator

	3.3 Air Filter 3.4 Lubricator 3.5 Check Valve 3.6 Silencer (Muffler) 3.7 Shut-off Valve
4. Signal Types	4.1 Analog 4.2 Digital 4.3 Electrical 4.4 Pneumatic
5. Technical Documentations	5.1 Circuit diagram 5.2 Data sheets 5.3 Manuals
6. Sequential Control	6.1 $A+ \rightarrow B+ \rightarrow C+ \rightarrow A- \rightarrow B- \rightarrow C-$ 6.2 $A+ \rightarrow B+ \rightarrow C+ \rightarrow C- \rightarrow B- \rightarrow A-$ 6.3 $A+ \rightarrow B+ \rightarrow A- \rightarrow C+ \rightarrow B- \rightarrow C-$ 6.4 $A+ \rightarrow B+ \rightarrow C+ \rightarrow B- \rightarrow A- \rightarrow C-$ 6.5 $A+ B+ C+ \rightarrow A- \rightarrow B- \rightarrow C-$
7. Appropriate Devices	7.1 Relay 7.2 Magnetic contactor 7.3 Coils, 7.4 Timers, 7.5 Counters

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions and specifications of PLC and electro-pneumatic components. 1.2 Fundamentals of PLC programming and ladder logic principles. 1.3 Safety standards and regulatory requirements related to electro-pneumatic systems. 1.4 Control strategies used in electro-pneumatic applications. 1.5 Methods for translating control requirements into ladder logic programs. 1.6 Simulation techniques for ladder logic. 1.7 Electrical and pneumatic interface requirements and standards. 1.8 Signal types and their matching with PLC inputs and outputs. 1.9 Communication protocols between PLC and electro-pneumatic components. 1.10 Sequential control theory and its application in electro-pneumatic systems. 1.11 Programming techniques for multi-cylinder coordination and
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	interlocking. 1.12 Timing, delay, compare, latching, flip-flop and counting functions in sequential programming.
2. Underpinning Skills	2.1 Identifying PLC and electro-pneumatic components. 2.2 Interpreting technical manuals and wiring diagrams. 2.3 Recognizing communication protocols and signal types used for PLC to electro-pneumatic system interaction. 2.4 Using PLC programming software to develop and simulate programs. 2.5 Connecting electro-pneumatic devices to PLC inputs/outputs. 2.6 Using testing equipment to check interface connections. 2.7 Simulating and testing sequential control programs for multi-cylinder operations. 2.8 Integrating sensor feedback into control logic. 2.9 Optimizing timing and control sequences for efficiency and safety.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified components of PLC based electro- pneumatic control system 1.2 interpreted and develop PLC ladder logic for electro-pneumatic applications 1.3 interfaced electro-pneumatic components with PLC 1.4 programed PLC for sequential control
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2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: UNDERSTAND CONCEPT OF HYDRAULIC CONTROL SYSTEM	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LE-PHS-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand concept of hydraulic control system. It specifically includes the tasks of identifying components of hydraulic system, interpreting hydraulic circuit diagrams and demonstrating safe practice in handling hydraulic system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Hydraulic System Components	1.1 <u>Major hydraulic components</u> are recognized. 1.2 Functions of major hydraulic components are described. 1.3 Symbols of hydraulic components are interpreted. 1.4 Input, Output ports of <u>valves</u> and <u>actuators</u> are located and their purpose are explained. 1.5 Components are differentiated by <u>operational type</u> .
2. Interpret Hydraulic Circuit Diagrams	2.1 Components of hydraulic circuit are identified. 2.2 Functions of individual hydraulic components represented in the diagram are explained. 2.3 The flow of liquid within the circuit is traced and described. 2.4 Relationships between different circuit elements and their operational sequence are interpreted. 2.5 Troubleshooting points are identified based on the circuit diagram interpretation.
3. Demonstrate Safe Practice in Handling Hydraulic System	3.1 Safety procedures and guidelines are followed. 3.2 Hydraulic components and tools are inspected and liquid pressure is maintained. 3.3 Hydraulic circuit simulation is demonstrated. 3.4 Hydraulic application is demonstrated on training board. 3.5 Hydraulic system components are handled and operated according to safety standards and manufacturer instructions. 3.6 <u>Potential risks and hazards</u> are identified and controlled effectively to prevent accidents. 3.7 Unsafe conditions or practices are reported promptly.

Range of Variables

Variable	Range (May includes but not limited to):
1. Major Hydraulic Components	1.1 Liquid tank 1.2 Actuators (single acting and double acting cylinders) 1.3 DCV (Directional control Valve) 1.4 FCV (Flow control Valve) 1.5 Shutoff valve
2. Valves	2.1 2/2, 3/2, 4/2, 4/3, 5/2, 5/3-way DCV 2.2 1 way and 2-way flow control Valve 2.3 Normally open and normally closed push Valve.
3. Actuator	3.1 Single acting cylinder 3.2 Double acting cylinder
4. Operational Type	4.1 Manual, 4.2 Mechanical 4.3 Solenoid
5. Potential Risks and Hazards	5.1 High Pressure Release 5.2 Component Failure 5.3 Noise Exposure 5.4 Pinch/Crush Injuries 5.5 Incorrect Connections 5.6 Electrical Hazards

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic principles and functions of hydraulic components. 1.2 Standard symbols for hydraulic components. 1.3 Purposes of common hydraulic components. 1.4 Understand circuit diagrams and principles of hydraulic control circuits. 1.5 Functional relationships between circuit components. 1.6 Methods of troubleshooting circuit diagrams. 1.7 Hazards and risks associated with hydraulic systems. 1.8 Safe handling, installation, and maintenance procedures.
2. Underpinning Skills	2.1 Identifying and distinguishing hydraulic components visually. 2.2 Applying knowledge of component function to practical identification. 2.3 Demonstrating simulation of hydraulic circuit. 2.4 Demonstrating hydraulic application on training board. 2.5 Tracing flow paths and control logic from diagrams. 2.6 Operating circuit components and identifying

	troubleshooting points and faults. 2.7 Implementing safe manipulation and assembly of hydraulic components. 2.8 Identifying and mitigating of hazards. 2.9 Executing emergency shutdown and isolation procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified components of hydraulic system 1.2 interpreted hydraulic circuit diagrams. 1.3 demonstrated safe practice in handling hydraulic system
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ELECTRO- HYDRAULIC CONTROL APPLICATION	Nominal Duration: 30 Hrs.	Unit Code: SICIP-LE-PHS-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to use electro- hydraulic control application. It specifically includes the tasks of identifying electro-hydraulic control circuits, interpreting logical control functions, assembling electro-hydraulic circuits, simulating and testing electro-hydraulic system and applying electro-hydraulic control application circuits in training board.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Electro-Hydraulic Control Circuits	1.1 Standard symbols for electro-hydraulic components are recognized and identified. 1.2 Functions of each electro-hydraulic components within the circuit are explained. 1.3 Electrical and hydraulic parts of the circuit are distinguished and categorized. 1.4 Functions of <u>sensors</u> , actuators, <u>valves</u> , and <u>controllers</u> in the circuit is identified. 1.5 Circuit elements are matched to their respective control functions within the system.
2. Interpret Logical Control Functions	2.1 Relay based logical operations and functions used in control systems are identified and described. 2.2 Input and output relationships within the logical control system are analyzed and explained. 2.3 Logical sequences and conditions of cylinder movement are interpreted. 2.4 The effect of each logical function on actuators is determined and articulated. 2.5 Faults or irregularities in sequence control are detected through analysis. 2.6 The overall sequential control logic is interpreted.
3. Assemble Electro-Hydraulic Circuits	3.1 Required electro-hydraulic components are selected and arranged within FluidSIM software according to design. 3.2 Circuit diagrams and simulation instructions are interpreted before assembly. 3.3 Electro-hydraulic components are connected correctly in FluidSIM following the circuit layout and operational

	logic. 3.4 Electrical connections and hydraulic lines are configured and linked using FluidSIM tools.
4. Simulate and Test Electro-Hydraulic System	4.1 Simulation settings and parameters in FluidSIM are configured to match system operational requirements. 4.2 The assembled electro-hydraulic circuit is simulated in FluidSIM to replicate real-world conditions. 4.3 System responses and performance data are monitored and recorded during the FluidSIM simulation. 4.4 Any faults or errors detected during the FluidSIM simulation are identified and documented. 4.5 Necessary corrections and adjustments are made within FluidSIM to address identified issues.
5. Apply Electro-Hydraulic Control Circuits	5.1 Electro-hydraulic control circuits for two-cylinder <u>sequences</u> are selected. 5.2 Cylinders A and B along with corresponding sensors, are correctly identified and integrated into the control circuit on the training board. 5.3 Hydraulic tubing and electrical wiring are connected. 5.4 Sensor inputs and cylinder movements are monitored to ensure synchronization. 5.5 Faults or discrepancies in cylinder sequences or sensor feedback are identified and corrected. 5.6 Safety and operational procedures are followed throughout the setup, testing, and shutdown.

Range of Variables

Variable	Range (May includes but not limited to):
1. Sensors	1.1 Solenoid Proximity Sensor 1.2 Magnetic Proximity Sensor 1.3 Capacitive Proximity Sensor 1.4 Optical Proximity Sensor
2. Valves	2.1 4/2way solenoid valve 2.2 4/3way solenoid valve
3. Controllers	3.1 Relay based 3.2 PLC based
4. Sequences	4.1 A+A-B+B- 4.2 A+B+A-B- 4.3 A+B+B-A-

Curricular Content Guide

1. Underpinning Knowledge	1.1 Standard symbols used in electro-hydraulic circuit diagrams. 1.2 Functions of key electro-hydraulic components. 1.3 Electrical and pneumatic parts of the circuit. 1.4 Fundamentals of logic operations and their implementation in control systems. 1.5 Use of relay logic in electro- hydraulic control. 1.6 Assembly procedures and best practices for electro-hydraulic hardware. 1.7 Simulation software tools of FluidSIM and their features. 1.8 Testing procedures and performance criteria for electro-hydraulic systems. 1.9 Fault detection and corrective actions during testing. 1.10 Configuration and setup of training boards for electro-hydraulic circuits. 1.11 Sensor and actuator integration on training platforms.
2. Underpinning Skills	2.1 Interpreting electro- hydraulic circuit diagrams. 2.2 Recognizing electrical and hydraulic parts of the circuit. 2.3 Analysing and understanding logic control sequences. 2.4 Translating logical functions into practical control applications. 2.5 Diagnosing and correcting logic faults. 2.6 Handling, connecting, and securing electro- hydraulic components. 2.7 Wiring electrical connections and routing hydraulic lines. 2.8 Operating simulation software to model electro- hydraulic circuits. 2.9 Setting up test parameters and monitoring system performance. 2.10 Configure and operate training board setups for electro-hydraulic control.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified electro- hydraulic control circuits 1.2 interpreted logical control functions 1.3 assembled electro- hydraulic circuits 1.4 simulated and tested electro- hydraulic system 1.5 applied electro- hydraulic control application circuits in training board
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ADVANCED ELECTRO-HYDRAULIC CONTROL APPLICATION	Nominal Duration: 35 Hrs.	Unit Code: SICIP-LE-PHS-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform advanced electro-hydraulic control application. It specifically includes the tasks of identifying components of PLC based electro-hydraulic control system, interpreting and developing PLC ladder logic for electro-hydraulic applications, interfacing electro-hydraulic components with PLC and programming PLC for sequential control.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Components of PLC based Electro-Hydraulic Control System	1.1 <u>Electro- Hydraulic system components</u> are recognized and described. 1.2 Integration interfaces between PLC and electro- hydraulic components are identified and explained. 1.3 Communication protocols and signal types used for PLC to electro- hydraulic system interaction are recognized. 1.4 Safety devices and <u>protective components</u> within the control system are identified and their functions explained. 1.5 <u>Technical documentations</u> are consulted and interpreted correctly for component identification.
2. Develop PLC Ladder Logic	2.1 PLC ladder logic diagrams are analyzed and interpreted to understand the required control sequences. 2.2 Electro-hydraulic system requirements are translated into ladder logic functions. 2.3 Ladder logic is developed using <u>appropriate devices</u> . 2.4 The ladder logic program is simulated or tested. 2.5 Programming errors and logic faults are detected and corrected. 2.6 Ladder logic programs are downloaded to the PLC hardware and integrated with electro- hydraulic systems.
3. Interface Electro-Hydraulic Components with PLC	3.1 <u>Electro-hydraulic components</u> are identified and prepared for PLC interfacing. 3.2 Electrical connections between PLC modules and electro-hydraulic devices are established. 3.3 <u>Signal types</u> from electro- hydraulic components are matched with PLC input/output specifications.

	3.4 <u>Communication protocols</u> for PLC and electro-hydraulic components are recognized and implemented. 3.5 Interface wiring and connections are tested for continuity, signal integrity, and correct operation. 3.6 Integration of PLC and electro- Hydraulic components is verified through functional testing.
4. Program PLC for Sequential Control	4.1 <u>Sequential control</u> requirements for 2-cylinder operations are analyzed and documented. 4.2 PLC ladder logic languages are used to develop control sequences for 2-cylinder operations. 4.3 Control logic is expanded to incorporate 2-cylinder sequential operations. 4.4 Timers, counters, and sensor inputs are integrated to manage the sequence. 4.5 The final PLC program is downloaded to the hardware and tested. 4.6 System operation is monitored to ensure the sequential control of cylinders is performed according to design.

Range of Variables

Variable	Range (May includes but not limited to):
1. Electro-Hydraulic System Components	3.3 Single acting cylinder 3.4 Double acting cylinder 3.5 Solenoid valves 3.6 Proximity Sensor 3.7 Flow control valve 3.8 Shutoff valve 3.9 Level gauge
2. Protective Components	2.1 Pressure Relief Valve 2.2 Pressure Regulator 2.3 Check Valve 2.4 Shut-off Valve
3. Technical Documentations	3.1 Circuit diagrams 3.2 Datasheets 3.3 Manuals
4. Signal Types	4.1 Analog 4.2 Digital 4.3 Electrical 4.4 Hydraulic
5. Appropriate Devices	5.1 Contactor 5.2 Coils 5.3 Timers 5.4 Counters

6. Sequential Control	6.1 A+A-B+B- 6.2 A+B+A-B- 6.3 A+B+B-A-
7. Communication Protocols	7.1 Profinet 7.2 Ethernet 7.3 Modbus

Curricular Content Guide

1. Underpinning Knowledge	1.1 Functions and specifications of PLC and electro-hydraulic components. 1.2 Fundamentals of PLC programming and ladder logic principles. 1.3 Safety standards and regulatory requirements related to electro- hydraulic systems. 1.4 Control strategies used in electro- hydraulic applications. 1.5 Methods for translating control requirements into ladder logic programs. 1.6 Simulation techniques for ladder logic. 1.7 Electrical and hydraulic interface requirements and standards. 1.8 Signal types and their matching with PLC inputs and outputs. 1.9 Communication protocols between PLC and electro-hydraulic components. 1.10 Sequential control theory and its application in electro-hydraulic systems. 1.11 Programming techniques for multi-cylinder coordination and interlocking. 1.12 Timing, delay, and counting functions in sequential programming.
2. Underpinning Skills	2.1 Identifying PLC and electro-pneumatic components. 2.2 Interpreting technical manuals and wiring diagrams. 2.3 Recognizing communication protocols and signal types used for PLC to electro-pneumatic system interaction. 2.4 Using PLC programming software to develop and simulate programs. 2.5 Connecting electro-pneumatic devices to PLC inputs/outputs. 2.6 Using testing equipment to check interface connections. 2.7 Simulating and testing sequential control programs for multi-cylinder operations. 2.8 Integrating sensor feedback into control logic. 2.9 Optimizing timing and control sequences for efficiency and safety.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified components of PLC based electro-hydraulic control system 1.2 interpreted and developed PLC ladder logic for electro-hydraulic applications 1.3 interfaced electro-hydraulic components with PLC 1.4 programed PLC for sequential control
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: TROUBLESHOOT AND MAINTAIN PNEUMATIC AND HYDRAULIC SYSTEM OPERATIONS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-LE-PHS-08-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to troubleshoot and maintain pneumatic and hydraulic system operations. It specifically includes the tasks of inspecting pneumatic and hydraulic system components, diagnosing faults in pneumatic and hydraulic systems, performing maintenance on pneumatic and hydraulic systems and testing and verifying system operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect Pneumatic and Hydraulic System Components	1.1 Pneumatic and hydraulic components are visually inspected for signs of wear, damage, and leakage. 1.2 System pressure, flow, and temperature are measured and recorded using appropriate instruments. 1.3 Component functionality is verified according to operational standards. 1.4 Safety and environmental compliance are ensured during inspection activities.
2. Diagnose Faults in pneumatic and hydraulic systems	2.1 Symptoms of system malfunctions are identified through observation and data analysis. 2.2 <u>Diagnostic tools</u> and techniques are applied to isolate faulty components or circuit sections. 2.3 Root causes of pneumatic and hydraulic <u>faults</u> are identified using systematic troubleshooting methods. 2.4 Fault diagnosis reports are prepared and communicated.
3. Perform pneumatic and hydraulic systems maintenance	3.1 <u>Preventive maintenance tasks</u> are carried out. 3.2 <u>Corrective maintenance</u> is performed to restore system functionality. 3.3 Pneumatic and hydraulic components are serviced or replaced. 3.4 Maintenance activities are documented, including parts replaced, tests conducted, and system status after maintenance.
4. Test and verify system operation	4.1 Pneumatic and hydraulic systems are tested under operational conditions to verify repair effectiveness. 4.2 <u>Performance parameters</u> are monitored and compared

	with standards.
	4.3 System safety and operational readiness are confirmed.

Range of Variables

Variable	Range (May includes but not limited to):
1. Diagnostic Tools	1.1 Pressure Gauges 1.2 Flow Meters 1.3 Temperature Sensors/Thermometers 1.4 Leak Detectors (Soap Solution) 1.5 Multimeters 1.6 Digital Pressure Sensors
2. Preventive Maintenance	2.1 Preparing monthly maintenance schedule 2.2 Components cleaning 2.3 Viscosity checking 2.4 Lubricating 2.5 Replacing faulty parts
3. Corrective Maintenance	3.1 Repairing liquid and air leakages 3.2 Restoring required pressure 3.3 Repairing actuators and valve
4. Faults	4.1 Leaks 4.2 Blockages 4.3 Pressure drops 4.4 Mechanical failures
5. Performance Parameters	5.1 Pressure 5.2 Flow rate 5.3 Response time

Curricular Content Guide

1. Underpinning Knowledge	1.1 Diagnostic and troubleshooting procedures. 1.2 Visually inspection technique of pneumatic and hydraulic components. 1.3 Manufacturer specifications and maintenance manuals. 1.4 Impact of environmental factors on system faults. 1.5 Fault rectification process. 1.6 Use and interpretation of diagnostic tools and testing instruments. 1.7 Types of maintenance: preventive, corrective, and predictive 1.8 Lubrication requirements and types of lubricants for components
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	1.9 Procedures for component removal, replacement, and adjustment 1.10 Use of hand and power tools specific to pneumatics and hydraulics. 1.11 Testing methods for pneumatic and hydraulic systems. 1.12 Performance criteria and operational parameters. 1.13 Safety precautions during system testing.
2. Underpinning Skills	2.1 Inspecting system components and connections visually. 2.2 Using inspection instruments and measuring devices. 2.3 Recognizing abnormal conditions and potential hazards. 2.4 Operating diagnostic tools. 2.5 Performing routine and scheduled maintenance tasks efficiently. 2.6 Handling and disposal of hydraulic fluids and lubricants safely. 2.7 Using of tools and equipment for maintenance and repair. 2.8 Conducting operational tests safely and effectively. 2.9 Identifying and rectifying any residual faults.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity. 4.4 Drawings, specifications and work instructions 4.5 Manuals.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 inspected pneumatic and hydraulic system components 1.2 diagnosed faults in pneumatic and hydraulic systems 1.3 performed maintenance on pneumatic and hydraulic systems
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	1.4 tested and verified system operation.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Unit of Competency

Workshop/Lab Facility Standard

Course Name:	Pneumatic & Hydraulic System Operation and Maintenance
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop with internet facility	25
2.	Multimedia projector & screen	02
3.	Pneumatic training board	10
4.	Hydraulic training board	10
5.	FluidSIM software	20
6.	PLC	20
7.	Single acting cylinders (pneumatic)	30
8.	Double acting cylinders (pneumatic)	30
9.	Single acting cylinders (hydraulic)	20
10.	Double acting cylinders (hydraulic)	20
11.	5/2 way DCV (Diretional control valve)	10
12.	3/2way push switch valve	10
13.	3/2way push switch valve (NC)	20
14.	5/2 way selector switch	10
15.	3/2 way selector switch	10
16.	3/2 way roller lever valve	10
17.	Pneumatic timer NO	10
18.	Pneumatic timer NC	10
19.	Multi-meter	10
20.	Tool box with all accessories	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Pneumatic & Hydraulic System Operation and Maintenance, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.